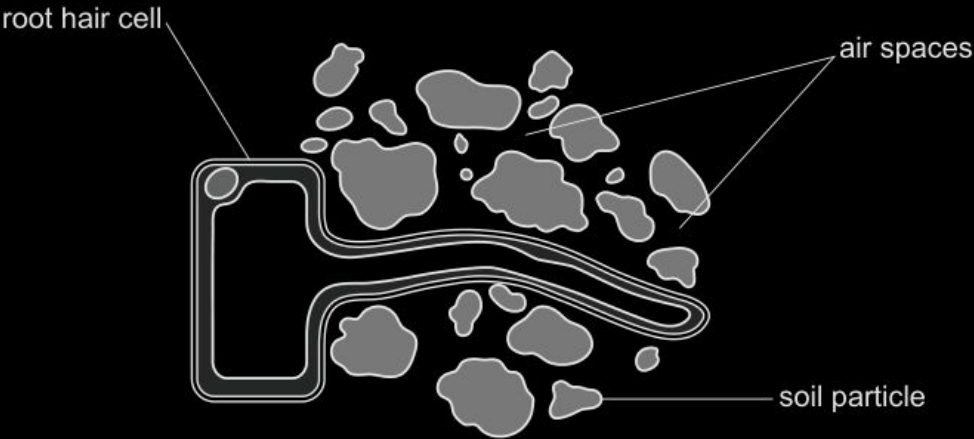


1 The diagram shows a root hair cell from a lemon tree and some soil particles.



(a) (i) State **one** feature, visible in the diagram, which shows that a root hair cell is a plant cell.
..... [1]

(ii) Describe how mineral ions pass from the soil into the root hair cell.
.....
.....
.....
.....
.....
..... [4]

(b) The lemon tree has grown in the same soil for many years so the soil now lacks magnesium.
Describe and explain the effect on the lemon tree of a lack of magnesium.
.....
.....
..... [2]

[Total: 7]

2 A person visits an eye doctor to have an eye test. On the wall of the doctor's room is a diagram of

1(a)(i)	cell wall or large / central / sap + vacuole ;	1	
1(a)(ii)	<i>any four from:</i> dissolved in water ; through the cell wall ; <u>carriers</u> in membrane ; through the cell / partially permeable AW + membrane ; by active transport / diffusion ; correct reference to concentration gradient ; using energy (for active transport) ; from (aerobic) respiration (for active transport) ;	4	
1(b)	yellowing / chlorosis / reduced growth / cannot photosynthesise / make glucose / starch / sugar / carbohydrate ; deficiency of chlorophyll ;	2	

7 (a) A root hair cell in a plant is respiring aerobically.

(i) Name the structures within the cell that carry out aerobic respiration.

..... [1]

(ii) Write the balanced symbol equation for the aerobic respiration of glucose, $C_6H_{12}O_6$.

..... $\longrightarrow$ [2]

(b) The plant experiences changing weather conditions including:

high temperature heavy rain high winds low humidity

State the weather condition that may lead to a root hair cell respiring anaerobically and explain your answer.

weather condition

explanation

..... [2]

[Total: 5]

8 (a) The number of individuals in a population can vary with time.

7(a)(i)	mitochondria / mitochondrion ;	1	
7(a)(ii)	$C_6H_{12}O_6 + 6O_2 ;$ $6H_2O + 6CO_2 ;$	2	A correct word equation for max 1 mark Ig any reference to energy or ATP
7(b)	heavy rain ; soil waterlogged / AW + no air spaces / oxygen available ;	2	

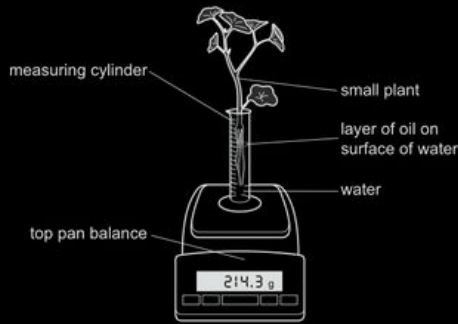
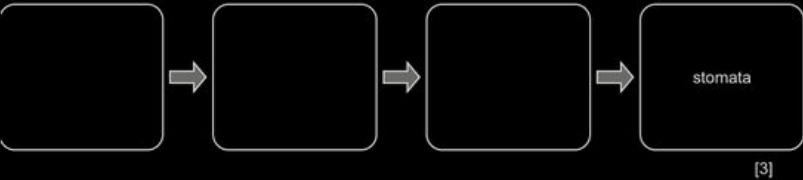


Fig. 1.1

- (a) (i) Name the process by which a plant loses water from cells to the atmosphere.
..... [1]
- (ii) Suggest the purpose of the layer of oil.
.....
..... [1]
- (iii) Describe **one** way in which a student could use this apparatus to obtain a value for the amount of water the plant loses in 24 hours.
.....
.....
..... [2]

- (b) Water is lost mainly through the plant's stomata.
Name **three** types of cell that the water passes through before it reaches the stomata in the leaves.
Write the names of the types of cell in the correct order in the boxes.



- (c) The experiment shown in Fig. 1.1 was repeated many times to investigate the effect of humidity on the rate of water loss.
All environmental conditions were kept constant except for humidity which was gradually increased.
Fig. 1.2 shows the axes for a graph.

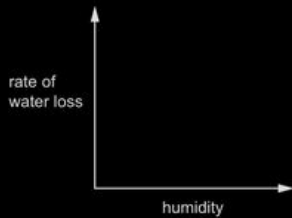


Fig. 1.2

- (i) Sketch a graph on these axes of the expected results for this investigation. [1]
- (ii) Explain what your graph shows about the effect of humidity on water loss.
.....
.....
.....
..... [3]

[Total: 11]

1(a)(i)	transpiration ;	1	
1(a)(ii)	prevent water loss / evaporation of water;	1	
1(a)(iii)	measure mass / volume at start + at end / after 24 hours ; subtract end value from start value / calculate difference ;	2	
1(b)	any three in correct sequence from: root hair (cell) ; (root) cortex (cell) ; xylem (vessel element) ; palisade / spongy / mesophyll (cell) ;	3	
1(c)(i)	line that is at its highest at a low humidity and at its lowest at a high humidity ;	1	
1(c)(ii)	the lower the humidity, the higher the rate of water loss AW ; water is lost by diffusion ; when humidity is low, the concentration of water outside will be lower than inside the leaf / atmosphere would not be saturated with water ;	3	

6 (a) (i) State what is meant by *transpiration*.

.....

.....

.....[3]

(ii) Explain how the transpiration stream is important for the production of proteins in a plant.

.....

.....

.....[2]

(b) Describe and explain how tissues in the stem of a plant become stained when a plant is carefully removed from the soil and placed in a solution of dye for a few hours.

.....

.....

.....

.....

.....

.....

.....[5]

[Total: 10]

7 (a) Describe the functions of the kidneys.





- 6 (a) (i) loss of water;
as vapour / evaporation;
from leaves / aerial parts of a plant / to the atmosphere;
through stomata (A between or through guard cells);

[max. 3]

- (ii) brings nitrates;
dissolved in water;
from soil / to combine with carbohydrates;

[max. 2]

- (b) dye diffuses (R ref. to osmosis);
through permeable (R semi- / partially- / selectively-);
(Ignore refs. to cell membrane)
(cell) wall;

of root (hair) cells;

xylem (any correct ref.);

ref. to travelling upwards (A refs. to transpiration pull / stream capillarity / root pressure AW);

[max. 5]

[Total: 10]

2 A potometer is used to measure water uptake by a plant. Fig. 2.1 shows the stem and flower of a plant in a potometer. As water is taken up, the bubble moves in the direction shown.

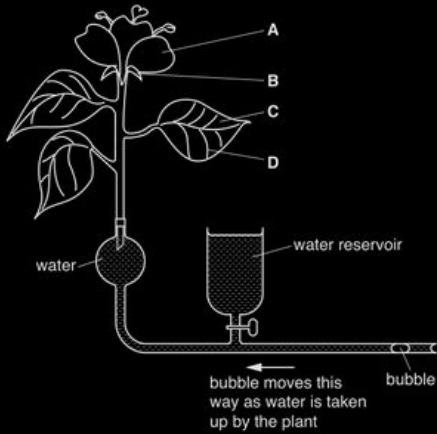
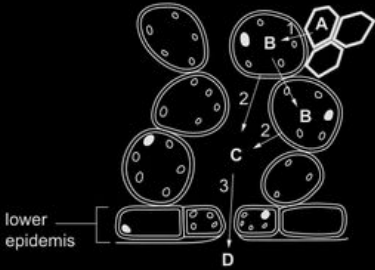


Fig. 2.1

- (a) Name the parts A, B, C and D in Fig. 2.1.
- A
- B
- C
- D [4]
- (b) Describe the pathway taken by water as it moves from the potometer, through the plant stem and into the surrounding air.
-
-
-
-
- [3]
- (c) (i) In an experiment, the bubble moved a distance of 60 mm in 10 minutes. Calculate the average rate at which the bubble moved in mm per min.
- mm per min
- [1]
- (ii) The experiment was repeated in an area of lower light intensity.
- Predict and explain what would happen to the rate at which the bubble moved.
- prediction.....
-
- explanation.....
-
-
- [3]
- [Total: 11]

2(a)	(A) petal ; (B) sepal ; (C) leaf / lamina / cuticle ; (D) vein / midrib / vascular (tissue) OR xylem + phloem ;	4	
2(b)	xylem ; to leaves / flower(s) ; between leaf cells ; surface of mesophyll cells ; (leaf) air spaces ; (through) stomata / guard cells ;	3	
2(c)(i)	6 ;	1	
2(c)(ii)	reduces / lower(s) AW ; less photosynthesis ; less water needed ; stomata / guard cells + open less / are closed ; less diffusion ; less evaporation / transpiration ;	3	
	Total:	11	

6 The diagram shows some cells from the lower part of a transpiring leaf.



- (a) (i) Identify cell type B.
- B [1]
- (ii) Water travels from A to D. The arrows on the diagram represent the movement of water molecules during transpiration. With reference to these arrows, name and define each process taking place.
- arrow 1
.....
.....
- arrow 2
.....
.....
- arrow 3
.....
..... [6]
- (b) Leaves from different plant species transpire at different rates in the same environmental conditions.
Suggest and explain structural differences in the lower surface of these leaves which might produce differences in the transpiration rate.
-
.....
.....
.....
..... [3]
- [Total: 10]

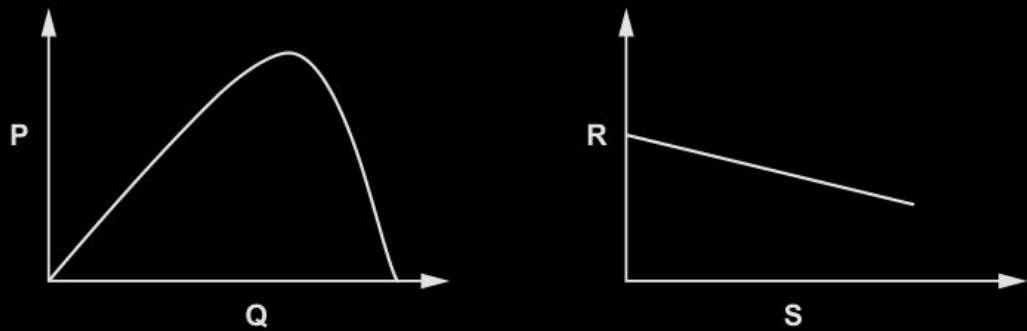
6(a)(i)	<u>spongy mesophyll</u> ;	1	
6(a)(ii)	<i>arrow 1</i> osmosis ; is the movement (of water molecules) from an area of higher water potential to lower water potential across a partially permeable membrane AW ; <i>arrow 2</i> evaporation ; is the change from liquid (water) to (water) vapour / gas ; <i>arrow 3</i> diffusion ; is the (net random) movement (of molecules) from a high to lower concentration / down a concentration gradient ;	6	A correct definition if linked to correct process but linked to the incorrect arrow.

6(b)	the surface area of the leaf ; the greater the surface area of the leaf, the greater the surface for evaporation ; the thickness of the cuticle ; the thicker the cuticle, the greater the barrier for water molecules ; the number / density of stomata ; as density of stomata increase, more places for water to escape from ; size of stomata ; as size of stomata increases, so does area from which water can be lost ; sunken stomata / ridged epidermis / curled up leaves ; sunken stomata / ridged epidermis / curled up leaves increase humidity ; hairs / projections ; hairs / projections increase humidity ;	3	Maximum two marks for structural features only with no explanation.
------	---	---	---

5 (a) Biological processes in plants are affected by environmental factors.

The graphs each show the effect of changing one environmental factor on a biological process.

The labels on the axes for each graph have been replaced with the letters **P**, **Q**, **R** and **S**.



Identify the correct axis label for each of **P**, **Q**, **R** and **S**.

Select from the following possible labels.

- | | | |
|------------------------|-------------------------------------|------------------------------|
| temperature | carbon dioxide concentration | rate of transpiration |
| light intensity | rate of photosynthesis | air humidity |

P

Q

R

S

[4]

(b) Describe the process of transpiration in the leaf of a plant.

.....

.....

.....

..... [3]

[Total: 7]

5(a)	(P) rate of photosynthesis ; (Q) temperature ; (R) rate of transpiration ; (S) air humidity ;	4	
5(b)	1 evaporation ; 2 surface of / cell wall / film of water / layer of water + mesophyll cells ; 3 water vapour ; 4 air spaces / intercellular spaces ; 5 diffusion / high to low concentration / down concentration gradient ; 6 water + through <u>stomata</u> ;	3	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

(a) Explain how the structure of a leaf is adapted for photosynthesis.

(b) Transpiration is both necessary for plants and a problem for plants.

Discuss this statement.

[Total: 10]

7(a)	1 large surface area + maximum light absorption ; 2 stomata + gaseous exchange / carbon dioxide in / oxygen out ; 3 thin + diffusion of gases / carbon dioxide in / oxygen out ; 4 air spaces / spongy mesophyll + diffusion of gases / gaseous exchange ; 5 chloroplasts / chlorophyll + <u>absorb</u> light ; 6 chloroplasts / chlorophyll + most dense in palisade (mesophyll) ; 7 palisade in upper leaf + receive most light ; 8 xylem (vessels) + bring water (from soil) ; 9 epidermal cells no chloroplasts / transparent + light reaches cells ; max 6	6	
7(b)	necessary (max 2): 1 provides (leaves with) water for photosynthesis ; 2 transports minerals / ions from roots / to leaves ; 3 provides plant cells with support / turgidity / turgor ; 4 cools the leaf / plant ; problem (max 2): 5 too much water is lost (through stomata / leaf) ; 6 (water loss) leads to wilting / death ; 7 plants may close stomata to prevent transpiration, which slows down photosynthesis ; max 4	4	

7 (a) (i) Explain how wilting occurs in a plant.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

(ii) Explain why mature plants do not wilt as much as newly-transplanted seedlings, even when both are exposed to similar conditions.

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

(b) Suggest an explanation for what would happen to a plant if it is accidentally ‘watered’ with a concentrated salt solution.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[3]

[Total: 10]

7	(a) (i)	water lost / plant dehydrated ; by transpiration / evaporation / through stomata ; faster than it is gained ; cells / tissues lose turgor / become flaccid / plasmolysed AW ; no longer support plant ; any condition leading to wilting ;	[max 4]	A lack of water, low humidity, high light intensity, high temperature, wind
	(ii)	(mature plant) root system undamaged ORA ; ref. root hairs ; water uptake sufficient for plant ORA ; plant cells / tissues turgid ORA ; xylem supports older plants / ref. lignin ;	[max 3]	I ref. more / less water uptake
	(b)	plant / leaves will wilt ; water lost ; from cells / tissues ; by osmosis / diffusion ; lower water potential in soil water / ref. correct water potential gradient AW ;	[max 3]	
			[Total: 10]	



- Suggest an explanation for this observation.

- Complete the equation below for photosynthesis using either words or symbols.



[1]

-
- The diagram illustrates the structure of a leaf. On the left, a whole leaf is shown with a central vein and lateral veins. A small section of the leaf is magnified on the right, showing a single mesophyll cell. The cell is roughly rectangular with a thick cell wall. Inside the cell, there is a large central vacuole (labeled P) that occupies most of the space. A nucleus (labeled Q) is located near the center. A chloroplast (labeled R) is situated near the cell wall, containing internal stacks of thylakoids.

Fig. 5.2

- P

Q

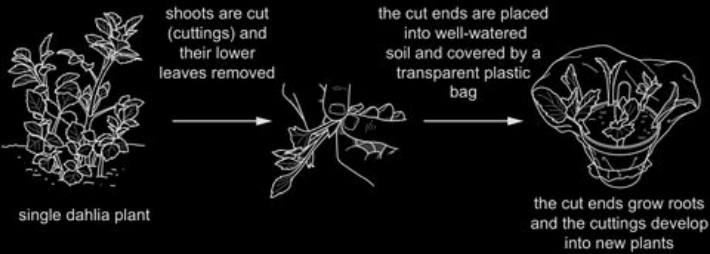
R

[3]

[Total: 10]

5 (a)	<i>wilting:</i> root hairs removed / roots damaged ; stem / xylem damaged ; less water absorbed ; plant loses water / transpires ; cells lose water / cells become flaccid ; <i>recovery:</i> root (hairs) regrow ; (more) water absorbed ; cells gain water again / become turgid ;	[max. 4]	R no water absorbed allow max.3 marks for <i>wilting</i> answers
(b)	carbon dioxide + water → oxygen ;	[1]	accept words or chemical formulae or a correctly balanced symbol equation I light / chlorophyll
(c) (i)	P = (lower) epidermal cell / epidermis / epidermal layer ; Q = guard cell ; R = stoma / stomata ;	[3]	
(ii)	fewer / no stomata / guard cells ; reduce / stop water loss / transpiration ; (appears) shiny / bright / ref. to cuticle ;	[max. 2]	
		[Total: 10]	

3 A gardener grows a species of dahlia, a flowering plant.
Each year the gardener chooses his best plants and takes cuttings to make new plants.



(a) (i) Removing the lower leaves and placing the cuttings in a transparent bag helps prevent them wilting until the roots develop.

Suggest and explain how these two steps help to prevent wilting.

removal of lower leaves

.....

.....

covering with transparent bag

.....

.....

[3]

(ii) Taking cuttings is a commercial application of asexual reproduction.

Define asexual reproduction.

.....

..... [2]

(b) The stems of the dahlia cuttings contain xylem vessels.

Describe **two** functions of xylem vessels in young stems of dahlias.
For each function explain **one** way in which xylem structure is adapted for this function.

function 1

.....

adaptation

.....

function 2

.....

adaptation

.....

[4]

(c) The gardener's dahlia cuttings grow into full-sized plants with flowers.

The gardener decides that he would like to produce a plant that combines the disease-resistant characteristics of one of these plants with the red petal colour of another.

With reference to named parts of the flowers, describe how he could achieve this.

.....

.....

.....

.....

.....

.....

.....

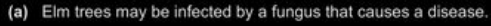
..... [5]

[Total: 14]

3(a)(i)	removal of leaves: reduction + surface area / stomata ;	1	
	covering with transparent bag: increase humidity ;	1	
	to reduce transpiration OR evaporation / water loss ;	1	
3(a)(ii)	any two from: production of <u>genetically identical</u> offspring / <u>clones</u> ; mitosis ; from one parent/ AW ;	2	

3(b)	function: xylem vessels transport water and/or minerals ;	1	
	adaptation: long / thin / no end cell walls / hollow tubes / no cytoplasm ;	1	
	function: (mechanical) support / keeps stem upright ;	1	
	adaptation: lignified OR woody / strengthened / hard + walls ;	1	
3(c)	any five from: 1 <u>artificial selection</u> / <u>selective breeding</u> ; 2 transfer pollen / pollination ; 3 from anther / stamen of one plant + e.g. red petal colour ; 4 to stigma / carpel / pistil of another ; 5 fertilisation OR gametes / genetic material + combined AW ; 6 grow seeds ; 7 select the ones with best combination of characteristics ;	5	

Question	Answer	Marks	Guidance
----------	--------	-------	----------



Explain how the response of the tree to infection leads to these symptoms of disease.

(i) Name the type of reproduction shown in the diagram.

4 Cardiopulmonary resuscitation (CPR) is a first aid procedure.

3(a)	1 hyphae / mycelium ; 2 saprotrophic / saprophytic / feeds on dead matter AW / decomposer ; 3 spores ; 4 cell wall + chitin ; 5 more than one nucleus / multinucleate ;	2	
------	---	----------	--

3(b)	1 less / no + water transported / water absorbed / transpiration ; 2 water loss greater than water uptake ; 3 less + water in cells ; 4 flaccid / loss of turgor / plasmolysed ; 5 magnesium / Mg ; 6 less / no + chlorophyll ;	5	
3(c)	1 transfer / carries / spreads AW + pathogen / fungus / parasite ; 2 one organism AW to another ; 3 not itself + affected / infected ;	2	
3(d)(i)	<u>asexual</u> ;	1	
3(d)(ii)	1 close together ; 2 connected AW ; 3 genetically identical / all genetically susceptible / clones ;	3	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

(b) A scientist measured the mass of water taken up and lost by a plant over 21 hours. The plant was kept at a constant temperature of 20 °C on a windowsill in a laboratory. Readings were taken every 90 minutes.

At 0 hours it was sunrise and at 15 hours it was sunset.

The results are shown in Fig. 2.2.

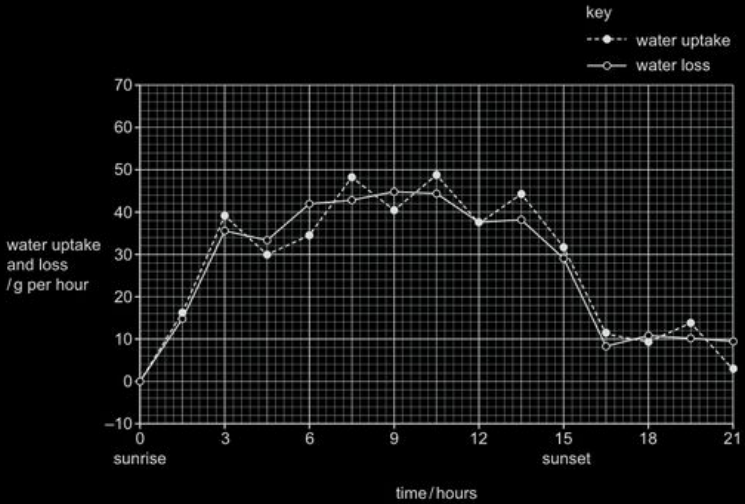


Fig. 2.2

(i) Plants lose water by evaporation from their leaves.

State the name of this process.

..... [1]

(ii) Explain the shape of the curve for **water loss** from the plant over the 21 hours of the experiment.

.....
.....
.....
.....
..... [2]

(iii) Explain why the shape of the curve for water loss is very similar to the shape of the curve for water uptake.

.....
.....
.....
.....
..... [3]

(iv) The scientist repeated the experiment. All conditions were kept the same except the temperature of the laboratory, which was 10 °C lower.

On Fig. 2.2, draw a new curve to show the expected results for **water loss** at the lower temperature. [1]

[Total: 12]

2(a)(i)	8 ;;	2	A 200 ÷ 2500 or 0.08 = 1 mark
2(a)(ii)	max three from: 1 homeostasis ; 2 maintaining constant body temperature ; 3 evaporation of sweat to remove heat (energy) ; 4 increased activity / increase in temperature means more sweat / ora ; 5 if volume lost in faeces / lungs / urine changes, then the percentage lost in sweat will change ;	3	
2(b)(i)	transpiration ;	1	

2(b)(ii)	max two from: increased water loss + during light / day / 0–15 hours / around sunrise + stomata open ; decreased water loss + during dark / night / 15–21 hours / around sunset + stomata close ; greater light intensity + faster evaporation / more diffusion / ora ;	2	
2(b)(iii)	max three from: 1 water evaporates / transpires from leaves ; 2 this reduces water potential of leaves ; 3 water moves up / through xylem vessels ; 4 transpiration pull / transpiration stream ; 5 forces of attraction between water ; 6 prevents loss of turgidity / prevents wilting ;	3	
2(b)(iv)	same shape of curve but below the original ;	1	

Question	Answer	Marks	Guidance
----------	--------	-------	----------

(a) Explain why phloem is more suitable for this purpose than any other plant tissue.

[3]

[7]

[Total: 10]

7 (a) Explain the differences between *mitosis* and *meiosis*. Your explanation should

- 6 (a) within reach;
contains sucrose / sugar / carbohydrate;
contains amino acids;
in solution / water;
not lignified / is softer **AW**;
R starch / glucose / mineral ions
Ig nutrients

[3]

- (b) impaired growth **AW** / death;
amino acids used to make protein;
less / no new cytoplasm;
less sucrose or carbohydrate;
less glucose;
for use in respiration;
energy used;
for any named purpose;
may introduce viruses / disease;
damage to plant **AW**;
Ig wilting

[max 7]

[Total: 10]

5990/22/M/J/24 Q6

6 Flowering plants are sometimes grown in soil to provide cut flowers that can then be displayed.

(a) A gardener watered flowering plants with a fertiliser solution containing dissolved chemicals.

Table 6.1 shows the information printed on the label of the container of fertiliser powder used to prepare this fertiliser solution.

Table 6.1

chemical component	mass/g per 100 g fertiliser powder
nitrates	4.0
phosphates	4.0
sulfur	1.0
magnesium	0.5

- (i) To prepare the fertiliser solution, the gardener dissolved 20 g of the fertiliser powder in 10 dm³ of water.
- Calculate the total mass of sulfur in the fertiliser solution prepared by the gardener.
- Space for working.

..... [2]

- (ii) Explain which chemical component of the fertiliser solution is the most important to plants for:
- making the leaves of the plants green in colour
-
-
- the production of enzymes by cells in the leaves of the plants.
-
- [4]

- (iii) The chemical components of the fertiliser solution are absorbed by the root hair cells of the plants and are then transported to the leaves.
- Complete the words below to name the three main tissues in the pathway taken by the chemical components of the fertiliser solution from the root hair cells to the leaves.
- The first letter of each word has been completed for you.
- C
- ↓
- X
- ↓
- M [3]

- (b) When flowers from the plants are cut and used for display, the leaves are removed before each stem is placed in water.
- A solution containing sucrose is often added to this water.
- Sucrose is **not** needed in fertiliser solution used to water plants growing in soil.
- Explain the benefit of providing sucrose to cut flowers.
-
-
-
-
- [3]

[Total: 12]

6(a)(i)	0.2 ; g ;	2	
6(a)(ii)	magnesium ; <u>chlorophyll</u> (makes leaves green); nitrates ; (enzymes made from) amino acids / proteins ;	4	
6(a)(iii)	(c)ortex ; (x)ylem ; (m)esophyll ;	3	
6(b)	1 no + photosynthesis ; 2 to produce glucose / glucose not produced ; 3 to produce sucrose / sucrose not produced ; 4 sucrose travels in phloem ; 5 for respiration / energy ; 6 flowers / plants + last longer / for days AW ; max 3	3	

1 Fig. 1.1 shows a plant with a magnified transverse section (TS) of the root and an incomplete, magnified transverse section (TS) of the stem.

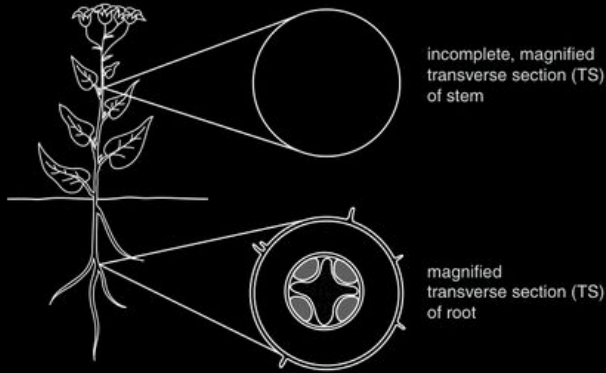


Fig. 1.1

- (a) The transverse section of the root in Fig. 1.1 shows the distribution of xylem and phloem tissue.
- (i) Using a labelling line and the letter **X**, show, on Fig. 1.1, the location of the xylem tissue in the transverse section (TS) of the root. [1]
 - (ii) Complete the transverse section (TS) of the stem in Fig. 1.1 to show the distribution of xylem and phloem tissue. [2]
 - (iii) Using a label line and the letter **P**, show the location of the phloem tissue that you have drawn. [1]

- (b) State the functions of phloem tissue.
-
-
-
-
-
-
- [3]

[Total: 7]

1 (a) (i)	Label line X on xylem on transverse section of root ;		[1]
(ii)	vascular bundle drawn towards outside of transverse section of stem ; vascular bundle divided into two sections ; three or more vascular bundles drawn ;		[max 2]
(iii)	label line P drawn in correct location (outer side of vascular bundle) ;		[1]
(b)	translocation ; transport ; dissolved / in solution ; sugar(s) / sucrose / products of photosynthesis ; amino acids ; reference from source + to sink / leaves to roots AW ;		[max 3]
[Total: 7]			

(b) List three structural changes that must occur in young, unmodified plant cells as they develop into xylem tissue.

- 1
- 2
- 3 [3]

(c) A small, leafy branch is cut from a tree. After some hours, the stem of the branch remains firm but the leaves become limp. Suggest an explanation for this.

stem remains firm
.....
.....
leaves become limp
.....
.....
..... [5]

[Total: 11]

4 Fig 4.1 shows a section through a ripe fruit of a tomato plant



- 3 (a) (i) cell wall + correct drawing (outside existing line);
- (ii) nucleus + correct drawing (must be in cytoplasm);
- (iii) vacuole / cytoplasm + correct drawing (if vacuole, must be larger than the nucleus.); [3]

If no marks scored through unacceptable drawings, allow max. 1 if all 3 are shown in the correct positions – vacuole in the middle, cell wall on the outside and nucleus between.

- (b) elongation of cell (R cylindrical);

loss of cell contents / or one named content (A ref. death of cell / hollow)
(R chloroplasts / dissolving of cell contents.);

loss of end walls;

addition of thickening / strengthening material / lignin; [max. 3]

- (c) (*stem*) contains xylem;

(xylem is) thickened / strengthened / contains lignin;

provides support;

(*leaf*) xylem less important for support;

loss of water / transpiration;

water not replaced / doesn't reach leaves;

cells become flaccid / lose turgor AW (R plasmolysed);

cells unable to support weight of leaf / wilts; [max. 5]

(a) similarities

[4]

[6]

Section C

7	(a)	<i>structural similarities:</i> long / elongated ; providing large surface area ; <i>functional similarities:</i> absorption / uptake ; active transport / diffusion ; ions / salts / minerals / named ; water ;	[max. 4]	
	(b)	<i>structural differences:</i> cell wall / no cell wall ; (root hair) part of one cell ; (villi) many cells / multicellular ; ref. absence of blood vessels / lacteals / ORA ; AVP ; <i>functional differences:</i> root hairs + absorb from the soil ; villi + absorb from the gut ; villi + absorb amino acids ; villi + absorb glucose ; villi + absorb lipids / glycerol / fatty acids ;	[max. 6]	
			[Total: 10]	

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[8]

[2]

[Total: 10]

9 (a) toxic / poisonous materials (R harmful);

waste products + metabolism AW;
(R refs. to salts)

urea;

liver / blood + kidneys;

urine;

urethra;

sweat;

skin / (epi)dermis;

carbon dioxide;

from cells / blood + alveoli;

exhaled / breathed out / expelled or removed from lungs AW; [max. 8]

(b) undigested food;

cellulose / fibre / lignin / roughage;

not from metabolism AW; [max. 2]

4 Living organisms excrete carbon dioxide.

(a) (i) Describe the meaning of the term excretion.

.....

.....

..... [2]

(ii) Name the process which produces carbon dioxide in living organisms and state how carbon dioxide is excreted by humans.

.....

.....

..... [2]

(b) The volume of carbon dioxide excreted varies over a 24-hour period and the factors that affect this are different for animals and plants.

Suggest and explain reasons why the volume of carbon dioxide excreted varies for both animals and plants.

animals

.....

.....

plants

.....

..... [4]

(c) Explain the effect of an increasing concentration of carbon dioxide in the atmosphere.

.....

.....

.....

..... [2]

[Total: 10]

5 The castor oil plant is a flowering plant which produces seeds.

4(b)	<i>animals</i> activities vary / example given ; so more respiration at some times ; <i>plants</i> respiration + constant / ongoing ; daylight + photosynthesis AW ; carbon dioxide used for photosynthesis ;	4 max two marks for plants
4(c)	any two from: acts as a greenhouse gas ; traps thermal / heat energy / global warming ; climate change / specific effect on land or organisms ;	2

7 (a) Explain why the lungs are considered to be organs of excretion.

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7 (a) CO₂ / water;

from body **AW**;

waste product / toxic material;

respiration;

metabolic (process);

[max 3]

(b) homeostasis / osmoregulation;

salts / ions / minerals / urea;

water;

from blood;

ref. reabsorption into blood;

in varying quantities / excess;

depending on food/water intake;

and on loss in sweating;

tissue fluid therefore at constant concentration;

water would otherwise enter or leave cells;

osmosis / diffusion;

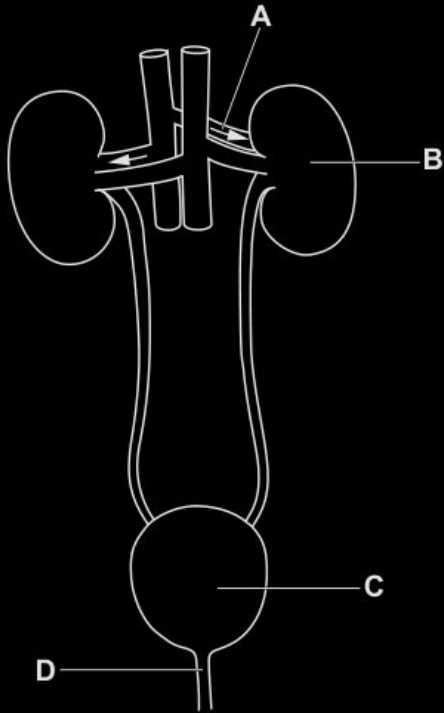
effect on cell appearance (e.g. swell / burst / shrink);

affect on enzyme action / metabolism;

[max 7]

[Total 10]

1 The diagram shows parts of the human body involved in excretion.



(a) Name the liquid that passes out of the human body through tube **D** and the waste product it contains.

liquid

waste product

[2]

(b) Describe how parts **A**, **B** and **C** are involved in removing the waste product that passes out of tube **D**.

.....
.....
.....
.....
..... [3]

[Total: 5]

2 Grapes are fruits that grow on a plant called a grape vine.

1(a)	urine ;	1	
	urea ;	1	A ammonia / excess water / excess minerals
1(b)	renal artery / A + brings urea / waste product to B / kidney AW ;	1	
	kidney / B + <u>filters</u> blood to remove urea / waste products ;	1	
	bladder / C + stores / keeps / collects <u>urine</u> ;	1	
Question	Answer	Marks	Guidance

9 (a) Compare the ureter and the urethra in terms of their functions in adult humans.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MS 5090/22/M/J/22 Q9

9(a)	1 both carry + urine / urea ; 2 ureter <u>from</u> kidney ; 3 ureter <u>to</u> bladder ; 4 urethra <u>from</u> bladder 5 urethra <u>out of</u> body ; (any two from) 6 male / penis / testis + urethra carries + semen / sperm ;	4	
9(b)	1 selectively / partially / semi + permeable ; 2 large surface area ; 3 fluid / dialysate ; 4 counter-current / continuously changed ; 5 urea / toxins / poisons / harmful substances + from blood / to fluid ; 6 water / salts / ions + between blood and fluid ; 7 diffusion / osmosis ; 8 high to low concentration / <u>down</u> concentration gradient ; 9 large <u>molecules</u> / <u>proteins</u> + stay in blood / do not enter fluid ;	6	

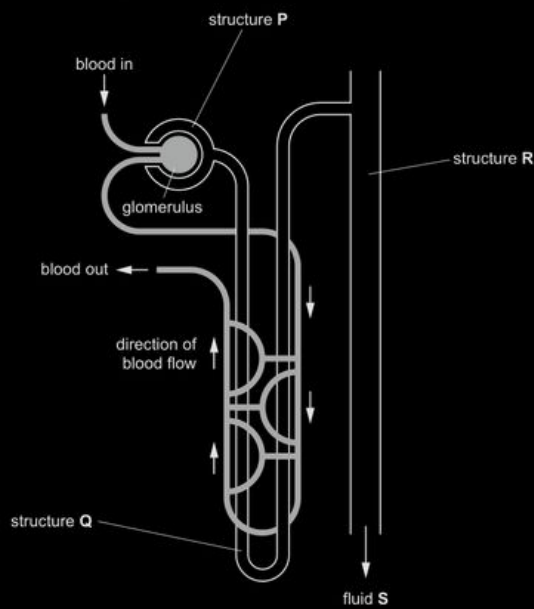


Fig. 4.1

- (a) (i) Identify each of the structures labelled P, Q and R in Fig. 4.1.
- P
- Q
- R [3]
- (ii) Name the organ which contains many nephrons.
- [1]
- (iii) Name the organ that stores fluid S before it is released from the body.
- [1]

- (b) Table 4.1 shows the composition of a sample of fluid S from a healthy person.
- The person that provided the sample of fluid S eats a balanced diet.

Table 4.1

component	concentration/arbitrary units
glucose	0.00
protein	0.00
ions	1.50
urea	2.00

- (i) State why it is important that urea is removed from the blood in fluid S.
- [1]
- (ii) Use your knowledge of the function of a nephron to explain the concentration of each of the following components of fluid S.
- glucose
-
-
-
-
- protein
-
-
- [5]
- (iii) Explain what would happen to the concentration of ions in fluid S if the person drank a large volume of water several hours before the sample was collected.
-
-
-
- [3]

Total: 141

4(a)(i)	(P) Bowman's capsule ; (Q) loop of Henle ; (R) collecting duct ;	3	
4(a)(ii)	kidney ;	1	
4(a)(iii)	(urinary) bladder ;	1	
4(b)(i)	toxic / toxin ;	1	
4(b)(ii)	<i>glucose</i> 1 enters Bowman's capsule / enters nephron / removed from blood AW ; 2 by filtration / filtered; 3 back into blood / reabsorbed + into blood ; 4 AVP ; <i>protein</i> 5 doesn't enter Bowman's capsule / doesn't enter nephron / no filtration / not removed from blood AW ; 6 protein + large AW ; max 5	5	
4(b)(iii)	ion concentration + decrease ; more water / higher volume of water / excess water + in fluid S / urine / excreted; homeostasis / to keep concentration of water in blood constant AW;	3	

Section C

7(a)	<i>any seven from:</i> transported in blood / plasma ; dissolved ; hepatic vein to vena cava / heart ; renal artery (to kidney) ; to kidney ; urea removed / filtered + from blood ; part of urine ; <u>ureter</u> (to bladder) ; bladder ; (bladder to) <u>urethra</u> ;	7	
7(b)	<i>any three from:</i> cycling / replenishing nitrogen ; used / converted by nitrifying bacteria / nitrogen fixing bacteria ; form + nitrates ; nitrates + taken up by plants ; production of proteins / amino acids ; <u>increased</u> soil fertility (for crops) / plants grow faster / bigger / AW ; animals / herbivores obtain protein / nitrogen from plants ;	3	

5090/21/01/14 02
2 Fig. 2.1 shows the digestion, absorption and assimilation of a chemical (B). It also shows the formation of its waste product, chemical F.

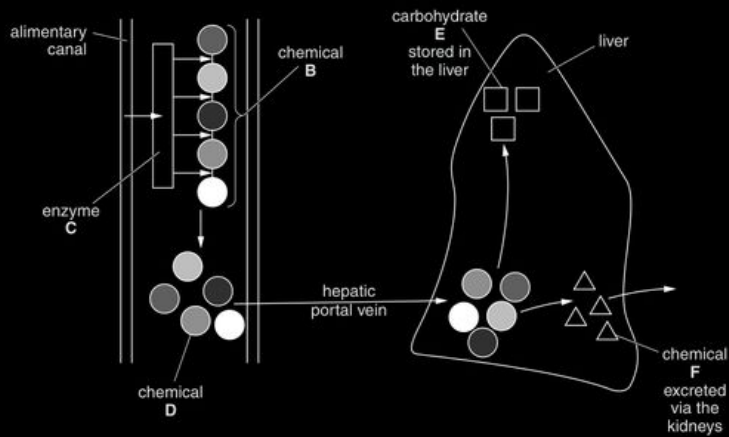


Fig. 2.1

(a) Name

chemical B

enzyme C

chemical D

carbohydrate E

chemical F [5]

(b) (i) Describe what happens to carbohydrate E before it can be used in metabolic reactions in body cells.

.....
.....
..... [2]

(ii) Name the process in which the product of carbohydrate E is used within a body cell.

process [1]

(iii) Describe how different types of body cells and tissues make use of this process.

.....
.....
.....
.....
.....
..... [4]

[Total: 12]

2	(a)	B = protein / (poly)peptide ; C = protease or named* ; D = amino acids* ; E = <u>glycogen</u> ; F = urea ;	[5]	*A ecf for incorrect substrate *A ecf for incorrect substrate R urine
	(b) (i)	broken down / converted / changed ; (role of) glucagon / adrenaline ; to <u>glucose</u> ; made soluble ;	[max. 2]	
	(ii)	<u>respiration</u> ;	[1]	

(b) Explain why carbon dioxide does not normally pass out of the leaves of a plant during the day.

[Total: 10]

9 (a) Describe and explain how a nitrate ion in the soil becomes part of a stored molecule in the

8(a)	digested / broken down /hydrolysis ; into amino acids ; by protease or named enzyme ; absorbed in small intestine, duodenum / ileum ; in villi / by capillaries / blood ; hepatic portal vein ; to liver ; deamination or described ; urea + in renal artery ; urea + taken to kidney ; blood filtered / urea leaves blood ; forms urine ; bladder / urethra ;	8	
------	--	---	--

8(b)	photosynthesis ; uses up CO ₂ ; faster than is made during respiration ;	2	
------	---	---	--

Question	Answer	Marks	Guidance
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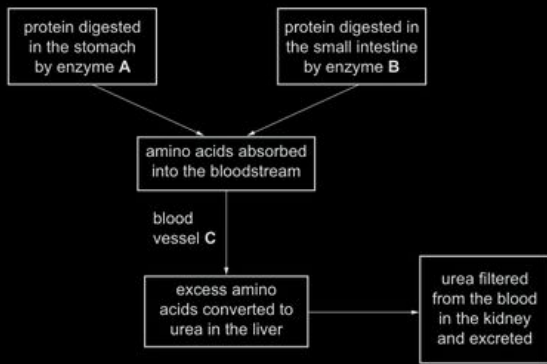


Fig. 3.1

Use Fig. 3.1 to answer these questions.

(i) State the name of enzyme A and enzyme B.

enzyme A

enzyme B

[2]

(ii) State the name of blood vessel C.

..... [1]

(iii) Name the capillary structures in the kidney that filter urea from the blood.

..... [1]

(b) Two groups of students, X and Y, investigate the effect of different diets on the concentration of urea in their urine.

For one day, the two groups eat different foods:

- group X eat bread, jam, orange juice, lettuce, fruit
- group Y eat nuts, milk, bread, cheese, meat.

Early the next morning, they collect and analyse their urine to measure the concentration of urea.

Table 3.1 shows their results.

Table 3.1

	concentration of urea/g per 100 cm ³ of urine						
	individual students						mean
group X	1.5,	1.7,	2.3,	1.6,	2.1,	1.4	1.8
group Y	1.8,	1.9,	2.3,	2.5,	1.9,	2.4	

(i) Calculate the mean concentration of urea for the students in group Y.

Give your answer to **two** significant figures.

Space for working.

mean concentration of urea =g per 100 cm³ of urine [2]

(ii) Explain the difference in the mean concentration of urea for the two groups of students.

.....
.....
.....
.....
.....
.....
..... [3]

(iii) One of the students said: 'To make a fair comparison, we should have made sure that the students in each group drank the same volume of water'.

Explain why this student is correct.

.....
.....
.....
.....
.....
..... [2]

3(a)(i)	pepsin ; trypsin ;	2	
3(a)(ii)	hepatic portal vein ;	1	
3(a)(iii)	glomerulus ;	1	
3(b)(i)	2.1 ;	2	A 12.8 / 6 OR 2.133 = 1 mark
3(b)(ii)	group Y ate more protein / ate protein rich foods ; so / in group Y, more amino acids are deaminated to produce urea ; so / in group Y, more urea is filtered / passes into the urine in the kidney ;	3	A ora for group X
3(b)(iii)	idea that the volume of water drunk, will affect the volume of water used to form urine ; this / the volume of water drunk, will affect the concentration of urea in the urine ;	2	

Question	Answer	Marks	Guidance
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3 Fig. 3.1 shows diagrams that represent the blood flow into and out of three major organs through blood vessels P to U.

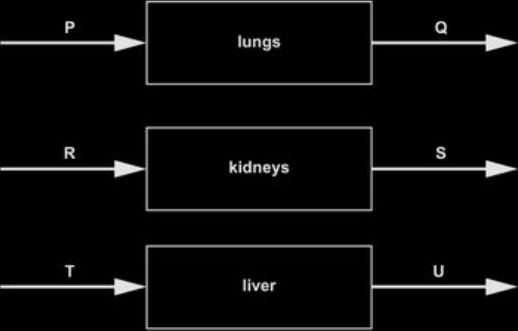


Fig. 3.1

(a) Identify each of the following blood vessels.

- Q [3]
R
U

(b) Compare the structures of blood vessels Q and R.

.....
.....
.....
.....
.....
..... [3]

(c) Explain each of the following.

(i) Blood in vessel S has a lower concentration of urea than blood in vessel R.

.....
.....
.....
.....
..... [3]

(ii) Blood in vessel U sometimes has a higher concentration of glucose than blood in vessel T.

.....
.....
.....
.....
..... [3]

[Total: 12]

4 Fig. 4.1 shows a Siamese fighting fish that lives in fresh water.

3(a)	(Q) <u>pulmonary vein</u> ; (R) <u>renal artery</u> ; (U) <u>hepatic vein</u> ;			3																
3(b)	<table border="1"><tr><td>Q – (pulmonary) vein</td><td></td><td>R – (renal) artery</td></tr><tr><td>thin + wall</td><td>+</td><td>thick + wall ;</td></tr><tr><td>thin muscle / thin elastic</td><td>+</td><td>thick muscle / thick elastic ;</td></tr><tr><td>wide + lumen AW</td><td>+</td><td>narrow + lumen AW ;</td></tr><tr><td>valves</td><td>+</td><td>no valves ;</td></tr></table>			Q – (pulmonary) vein		R – (renal) artery	thin + wall	+	thick + wall ;	thin muscle / thin elastic	+	thick muscle / thick elastic ;	wide + lumen AW	+	narrow + lumen AW ;	valves	+	no valves ;	3	/
Q – (pulmonary) vein		R – (renal) artery																		
thin + wall	+	thick + wall ;																		
thin muscle / thin elastic	+	thick muscle / thick elastic ;																		
wide + lumen AW	+	narrow + lumen AW ;																		
valves	+	no valves ;																		
3(c)(i)	1 urea + into kidney OR kidney removes + urea ; 2 filtration / filtering ; 3 nephron / Bowman's capsule / glomerulus ; 4 urea + excreted / in urine ;			3																

3(c)(ii)	1 (blood) glucose decreased / sugar low / body needs more sugar ; 2 glycogen ; 3 breakdown / hydrolysis + to glucose ; 4 glucose + passed out of liver / goes into U / hepatic vein / blood ; 5 (stimulated by) glucagon ;		
----------	--	--	--

Question	Answer	Marks	Guidance
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3 A man is cooking in a hot kitchen and sweat is being released by sweat glands in his skin.

(a) (i) Explain the term homeostasis with reference to sweating.

.....

.....

.....

..... [2]

(ii) Explain how receptors and blood vessels in the man's skin are also involved in homeostasis when he is in the hot kitchen.

.....

.....

.....

.....

.....

..... [4]

(b) Sweat contains very small quantities of the waste product urea.

State the name of the organ where most urea is made in the body.

..... [1]

(c) The table shows the concentrations of some chemicals found in the sweat, urine and blood plasma of a healthy human.

chemical	concentration of chemical /mmol per dm ³		
	sweat	urine	blood plasma
urea	22	393	6
sodium	66	110	141
chloride	59	103	99

(i) Use the data in the table to compare the chemical composition of urine and blood plasma.

.....

.....

.....

.....

.....

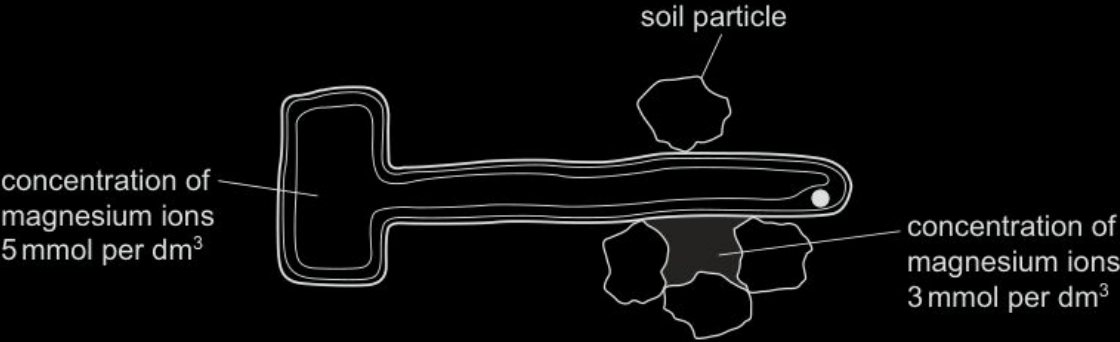
..... [4]

3(a)(i)	<i>any two from:</i> (homeostasis is) the maintenance of a <u>constant</u> ; <u>internal environment</u> / temperature ; return to norm / return to set-point / negative feedback ; <u>evaporation</u> of sweat removes heat / thermal energy / cools / decreases temperature ;	2	
3(a)(ii)	<i>any four from:</i> receptors + detection of temperature ; <u>impulse</u> to brain / CNS ; <u>vasodilation</u> ; arterioles / blood vessels widen ; increase in blood + near skin surface ; greater + loss of heat / thermal energy ; by radiation ;	4	
3(b)	<u>liver</u> ;	1	

3(c)(i)	<i>any four from:</i> less urea in blood plasma ; more sodium in blood plasma ; less chloride in blood plasma ; comparative data quote with units ; data manipulation, e.g. 50 × higher urea in urine than blood plasma ;	4	
3(c)(ii)	high(er) / over 22 (in patient) ; reduced filtration / reduced removal of urea (from kidney) ; reference to blood <u>plasma</u> ;	3	

Question	Answer	Marks	Guidance
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4 The diagram shows the concentration of magnesium ions in a root hair cell of a healthy plant and in the soil water surrounding the root hair cell. The plant continuously uses up magnesium ions.



For the plant to remain healthy, how will the magnesium ions move?

- A into the cell by active transport
- B into the cell by diffusion
- C out of the cell by active transport
- D out of the cell by diffusion

4	A	1
---	---	---

9 Which pathway is taken by water and mineral ions through a plant?

- A root hair cells → mesophyll cells → xylem → cortex cells
- B root hair cells → cortex cells → phloem → mesophyll cells
- C root hair cells → mesophyll cells → phloem → cortex cells
- D root hair cells → cortex cells → xylem → mesophyll cells

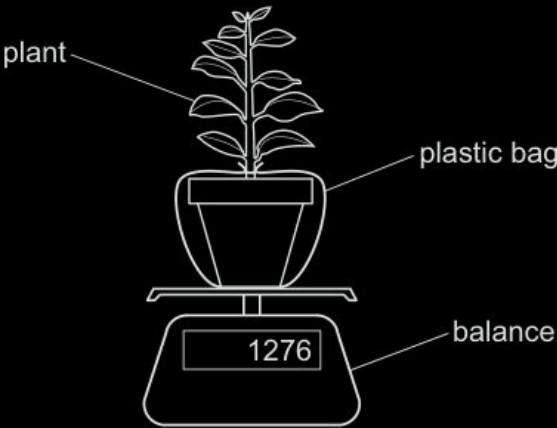
10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in

MS 5090/11/M/J/24 Q9

9	D	1
---	---	---

10 A student investigated the rate of transpiration in a green plant. The plant was growing in soil in a pot.

The student watered the plant and then put a plastic bag around the pot, leaving the leaves of the plant outside the bag. She measured the mass of the plant in the pot at the start and repeated this every two hours for six hours.



The results of the investigation are shown.

time of day	mass of pot and plant/g
09:00	1276
11:00	1270
13:00	1258
15:00	1252

What is the mean rate of transpiration over the six-hour period?

- A 43g/h
- B 12g/h
- C 8g/h
- D 4g/h

11 The diagram shows what happens to amino acids in part of the human body.

10	D	1
----	---	---

13 A healthy person eats a meal containing carbohydrates.

Two hours later, which structure will contain a liquid with an increased concentration of glucose?

- A colon
- B hepatic portal vein
- C renal artery
- D ureter

MS 5090/11/M/J/24 Q13

13	B	1
----	---	---

5090/11/M/J/24 Q22

22 Which row identifies the contents of urine in a healthy person?

	glucose	urea	water	ions
A	✓	x	✓	x
B	✓	✓	✓	✓
C	x	x	✓	✓
D	x	✓	✓	✓

MS 5090/11/M/J/24 Q22

22	D	1
----	---	---

5090/11/M/J/24 Q26

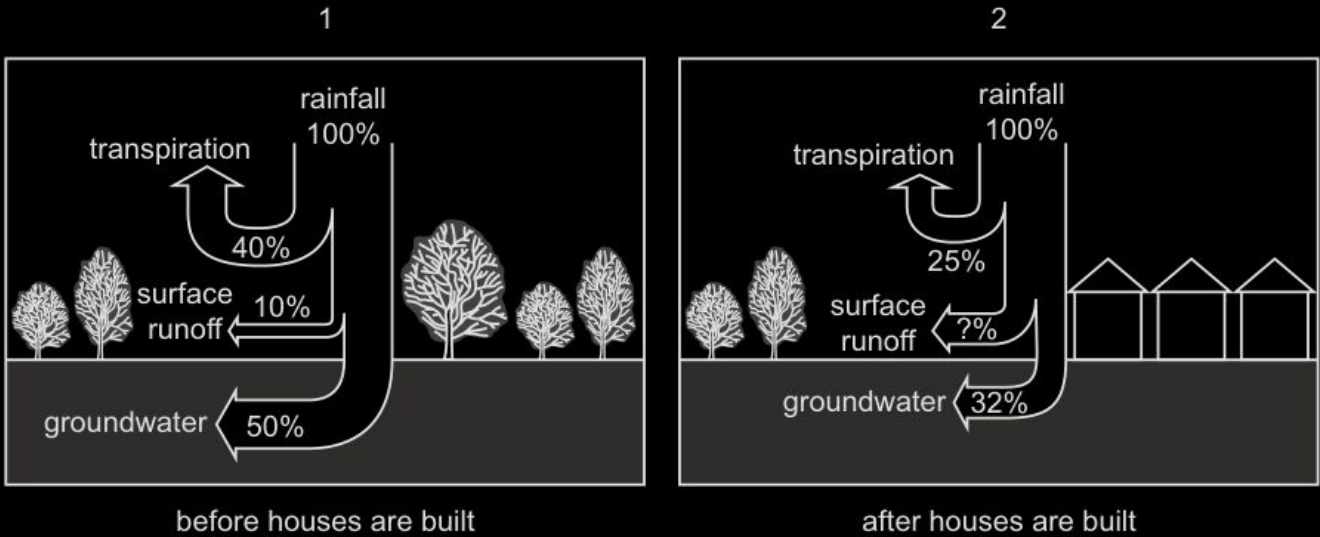
26 Which process is stimulated by adrenaline in the cells of the liver?

- A breakdown of glycogen, increasing the blood glucose level
- B breakdown of excess amino acids, forming urea
- C breakdown of proteins, releasing amino acids into the blood
- D conversion of excess blood glucose to glycogen

MS 5090/11/M/J/24 Q26

26	A	1
----	---	---

39 The diagrams show what happens to rain when it falls on an area of forested land before and after houses are built.



What is the difference in the percentage of surface runoff before and after building the houses?

- A increases by 33%
- B increases by 43%
- C decreases to 43%
- D decreases to 33%

39	A	1
----	---	---

9 Water moves from the soil to the atmosphere through a plant.

The water enters the plant through the root hair cells and moves into the root cortex.

Where does the water move into from the root cortex?

- A into the mesophyll cells
- B into the phloem cells
- C into the xylem vessels
- D through the stomata

10 The graph shows the loss of water vapour from two different plants growing in identical

MS 5090/12/M/J/24 Q9

9	C	1
---	---	---

5090/12/M/J/24 Q21

21 What is an example of excretion?

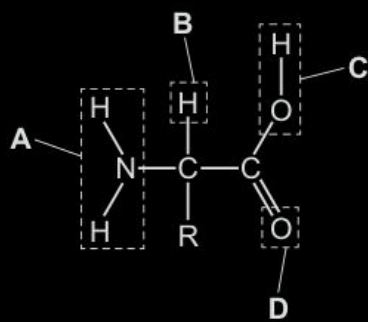
- A release of a hormone into the blood
- B removal of carbon dioxide from the lungs
- C removal of undigested food from the digestive system
- D release of water from the sweat glands

MS 5090/12/M/J/24 Q21

21	B	1
----	---	---

22 The diagram shows an amino acid molecule.

Which part of the amino acid molecule is removed and used to make urea?



23 What is a function of sensory neurones?

MS 5090/12/M/J/24 Q22

22	A	1
----	---	---

5090/12/M/J/24 Q38

38 Which process is involved in the conversion of ammonia to nitrates?

- A denitrification
- B excretion
- C nitrification
- D nitrogen fixation

39 Cutting down tropical rainforest trees has many consequences.

MS 5090/12/M/J/24 Q38

38	C	1
----	---	---

5090/11/O/N/24 Q4

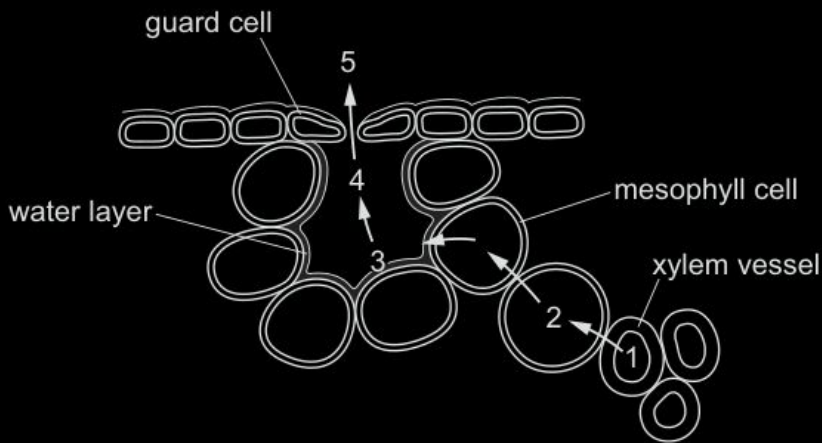
4 Which movement of a substance in a plant requires the plant to provide energy?

- A absorption of carbon dioxide by a palisade cell
- B absorption of oxygen by a mesophyll cell
- C nitrate uptake by root hair cells
- D transport of water up through the xylem

MS 5090/11/O/N/24 Q4

4	C	1
---	---	---

10 The diagram shows the movement of water through part of a leaf.



Which processes are involved in the movement of water at these stages?

	1–2	3–4	4–5
A	diffusion	evaporation	osmosis
B	evaporation	diffusion	osmosis
C	osmosis	diffusion	evaporation
D	osmosis	evaporation	diffusion

11 The diagram shows part of the human digestive svstem.

MS 5090/11/O/N/24 Q10

10	D	1
----	---	---

5090/11/O/N/24 Q15

15 Scientists have concluded that absorpion of mineral ions in plants requires energy from respiration.

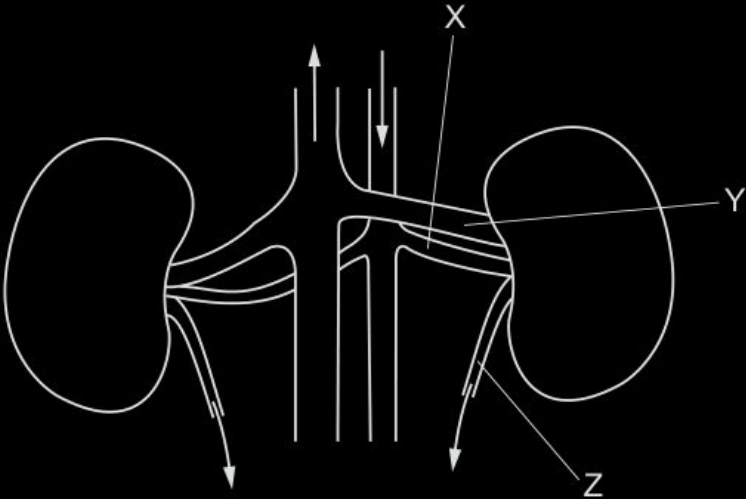
Which observation best supports this conclusion?

- A Carbohydrate is stored in the roots.
- B Living roots give off carbon dioxide.
- C The root hairs have a large surface area.
- D Uptake of mineral ions is reduced in lower oxygen concentrations.

MS 5090/11/O/N/24 Q15

15	D	1
----	---	---

22 The diagram shows part of the human urinary system.

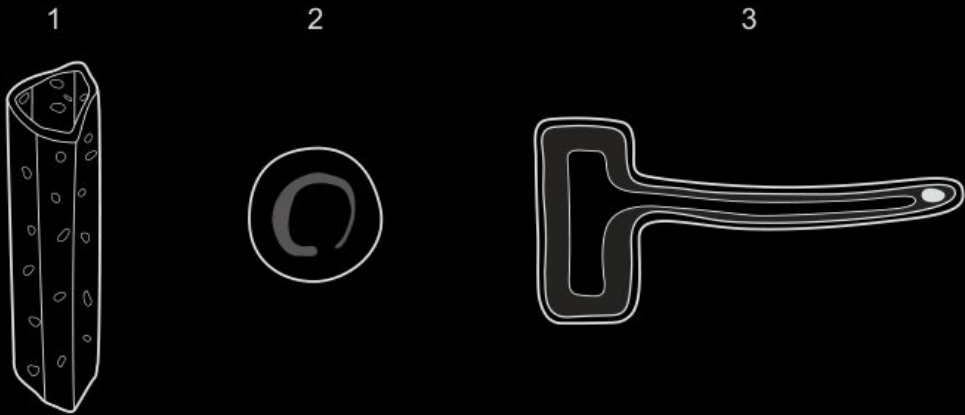


What are structures X, Y and Z?

	X	Y	Z
A	artery	vein	ureter
B	artery	vein	urethra
C	vein	artery	ureter
D	vein	artery	urethra

22	A	1
----	---	---

1 The diagram shows three different types of microscopic structure.



What are these structures?

	1	2	3
A	xylem vessel	root hair cell	red blood cell
B	red blood cell	xylem vessel	root hair cell
C	root hair cell	red blood cell	xylem vessel
D	xylem vessel	red blood cell	root hair cell

1	D	1
---	---	---

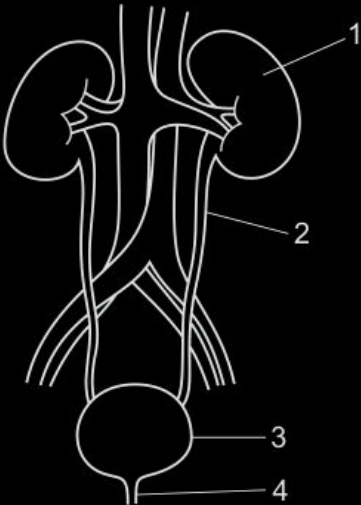
9 Root hair cells take in water and ions from the soil.

Which row shows how water and ions are taken into root hair cells?

	water	ions
A	active transport	active transport
B	active transport	osmosis
C	osmosis	osmosis
D	osmosis	active transport

9	D	1
---	---	---

22 The diagram shows the excretory system.

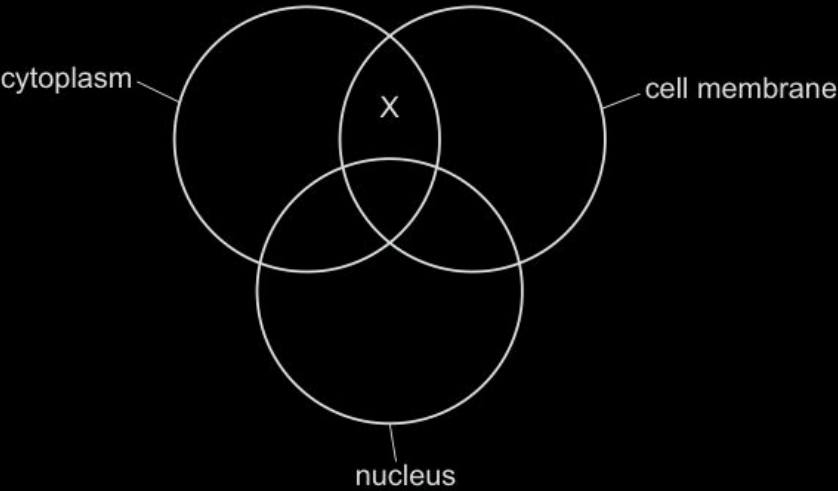


Which row identifies the names of the labelled structures?

	1	2	3	4
A	bladder	ureter	kidney	urethra
B	bladder	urethra	kidney	ureter
C	kidney	ureter	bladder	urethra
D	kidney	urethra	bladder	ureter

22	C	1
----	---	---

3 The diagram shows the structures found in cell X.

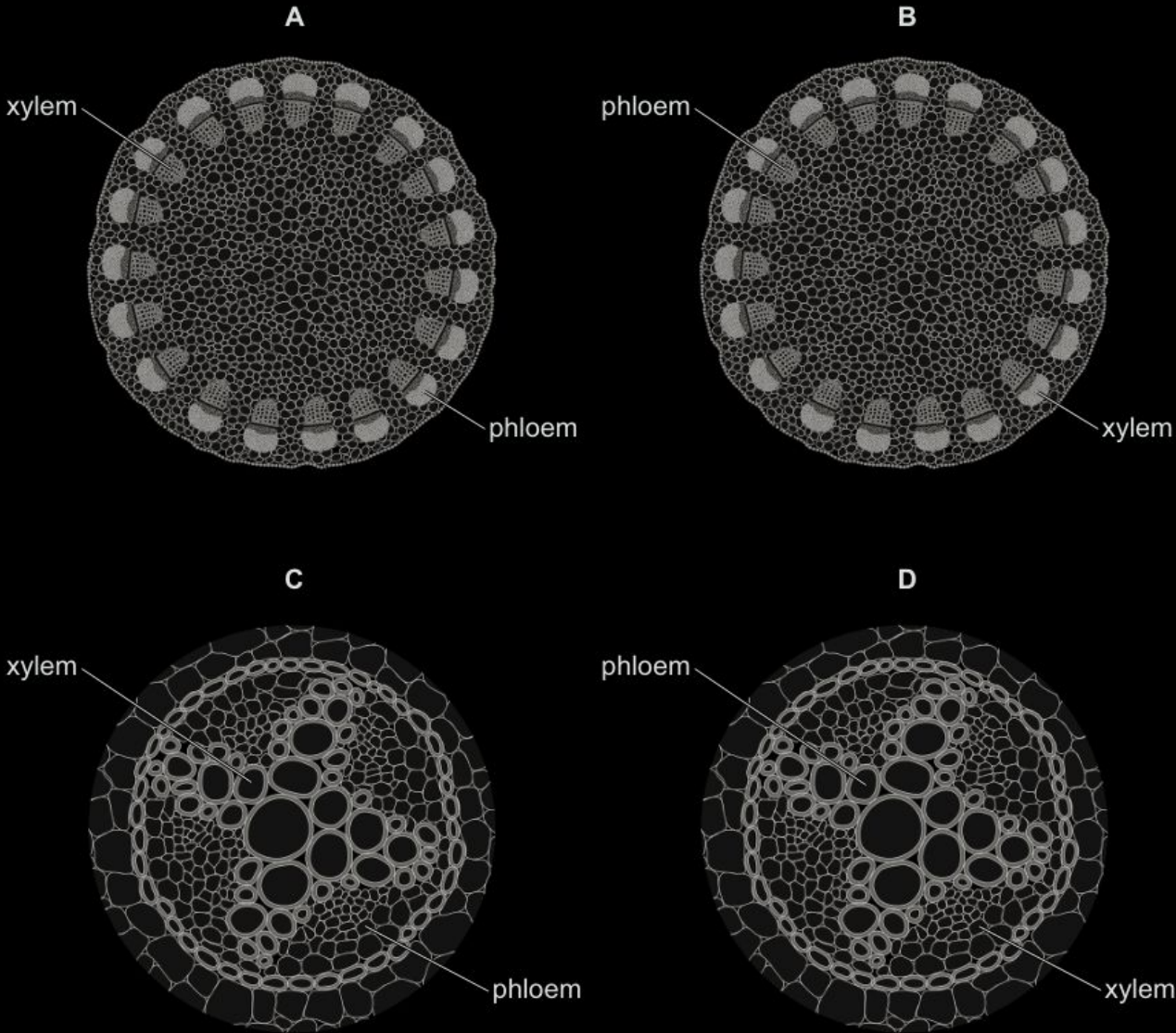


Which type of cell is X?

- A liver cell
- B red blood cell
- C root hair cell
- D xylem vessel

3	B	1
---	---	---

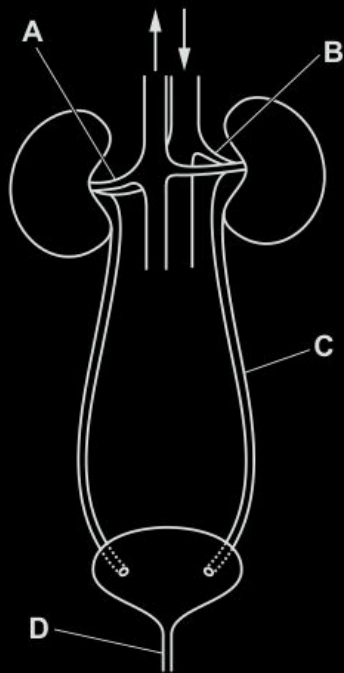
12 Which transverse section photomicrograph shows the position of the xylem and phloem in a non-woody dicotyledonous stem?



13 Students set up apparatus to measure the uptake of water by a leafy shoot.

12	A	1
----	---	---

23 Which structure is the ureter?



24 The processes of filtration and reabsorption take place in the kidneys.

MS 5090/11/M/J/25 Q23

23	C	1
----	---	---

24 The processes of filtration and reabsorption take place in the kidneys.

The daily amounts of some of the substances entering the kidneys in the blood and of these substances leaving the kidneys in the urine are shown.

substance	amount entering the kidneys in the blood	amount leaving the kidneys in the urine
water	150 dm ³	1.5 dm ³
glucose	150 g	0 g
urea	50 g	30 g

Which percentage of each substance is reabsorbed by the kidneys during a day?

	% of substance reabsorbed by kidneys		
	water	glucose	urea
A	99	100	40
B	148.5	150	20
C	0.99	1.0	0.4
D	1.0	0.0	60

24	A	1
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29 A nurse tests a sample of urine from a patient using Benedict's reagent. The sample tests positive.

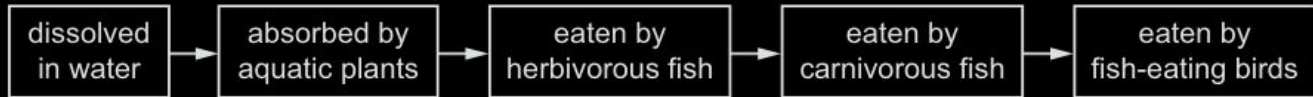
What is a possible explanation for this result?

- A The patient's liver is **not** deaminating protein efficiently.
- B The patient's pancreas is **not** producing sufficient insulin.
- C The patient has eaten too much glucose.
- D The patient's concentration of plasma proteins is too high.

29	B	1
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- 40** DDT is an insecticide that is harmful to living organisms. It builds up in the cells of living organisms as it is **not** excreted.

The diagram shows the path taken by DDT when it enters a lake.



Which organisms would the DDT harm the most?

- A** aquatic plants
- B** carnivorous fish
- C** herbivorous fish
- D** fish-eating birds

40	D	1
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5090/12/M/J/25 Q11

11 Glucose produced by photosynthesis can be converted into other carbohydrates or used immediately as glucose.

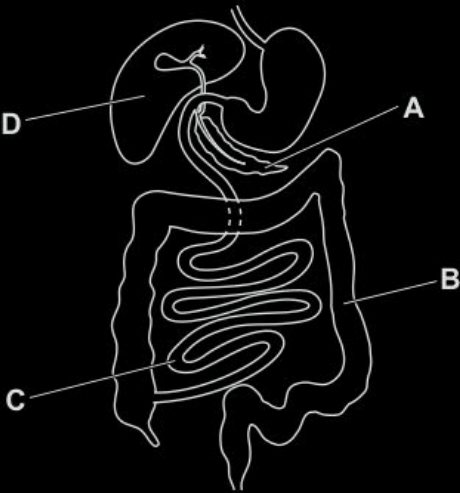
What is an immediate use of glucose in plants?

- A building cell walls
- B providing energy in respiration
- C storing energy
- D transporting energy in the phloem

MS 5090/12/M/J/25 Q11

11	B	1
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25 Which letter shows where urea is produced?



MS 5090/12/M/J/25 Q25

25	D	1
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5090/12/M/J/25 Q26

26 Which substance would **not** be present in the urine of a healthy person with a normal diet?

- A glucose
- B salts
- C urea
- D water

27 Which part of the eye contains the greatest density of light receptors?

MS 5090/12/M/J/25 Q26

26	A	1
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5090/11/O/N/25 Q5

5 Which process involves the movement of molecules against a concentration gradient?

- A active transport
- B diffusion
- C osmosis
- D transpiration

MS 5090/11/O/N/25 Q5

5	A	1
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14 Which processes use energy obtained from respiration?

- 1 contraction of intercostal muscles during breathing
- 2 nitrate ions entering a root hair cell against the concentration gradient
- 3 oxygen moving from the lungs into blood capillaries
- 4 production of antibodies by lymphocytes
- 5 water entering a bacterial cell by osmosis

A 1, 2 and 3 B 1, 2 and 4 C 2, 3 and 4 D 3, 4 and 5

MS 5090/11/O/N/25 Q14

14	B	1
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5090/11/O/N/25 Q15

15 During vigorous exercise, lactic acid builds up in the muscles. Blood leaving these muscles will have a high concentration of lactic acid.

Which process removes **most** of the lactic acid from the blood?

- A It is broken down in the liver.
- B It is excreted during sweating.
- C It is expired during faster breathing.
- D It is oxidised to release energy in muscles.

16 The table shows the mean rate of blood flow through some blood vessels.

MS 5090/11/O/N/25 Q15

15	A	1
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21 The table shows the concentration of five substances in the blood entering the kidney, in fluid entering the nephron and in the urine.

substance	concentration / mg per dm ³		
	blood entering the kidney	fluid entering the nephron	urine
urea	0.4	20.0	20.0
glucose	1.5	1.5	0.0
amino acids	0.8	0.8	0.0
ions	8.0	8.0	16.5
protein	82.0	0.0	0.0

Which substances are totally reabsorbed into the blood from the nephron?

- A amino acids and glucose
- B glucose and ions
- C protein and amino acids
- D protein and glucose

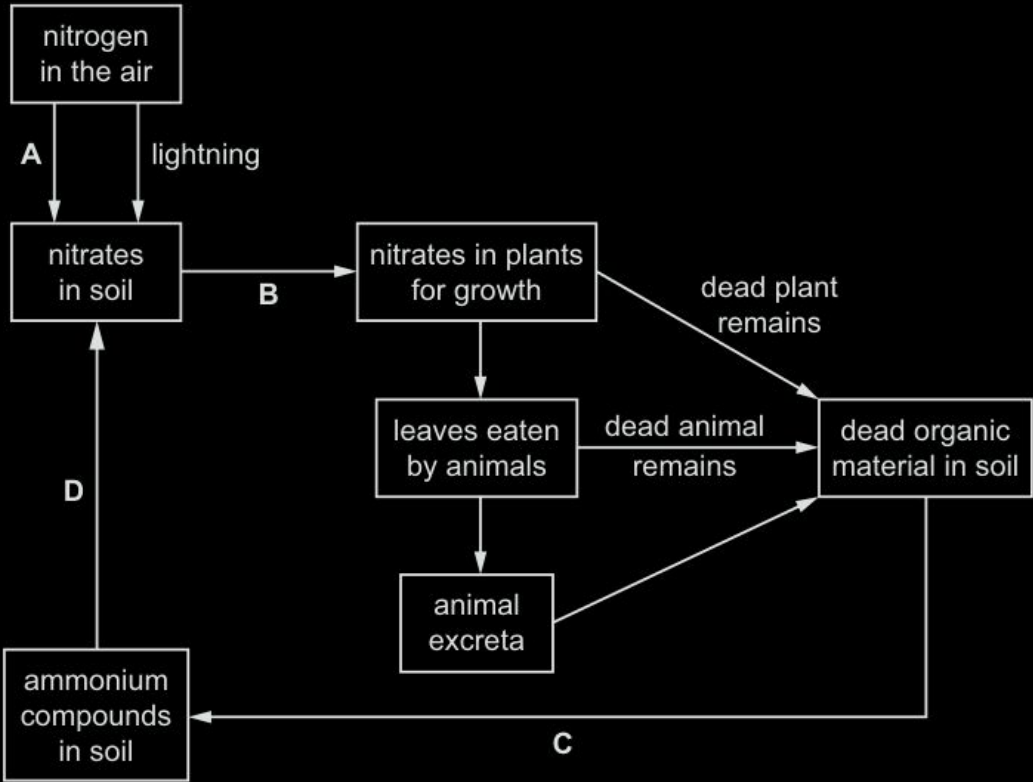
22 The graph shows blood glucose concentration over 6 hours in a day.

21	A	1
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39 Trees in a forest need a constant supply of nitrates to continue growing.

The diagram shows some of the stages of the nitrogen cycle that provide the necessary nitrates.

Which stage in this cycle is **not** brought about by living bacteria?



MS 5090/11/O/N/25 Q39

39	B	1
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5090/12/O/N/25 Q8

8 Oxygen is produced in leaf cells during photosynthesis and diffuses out of the leaf.

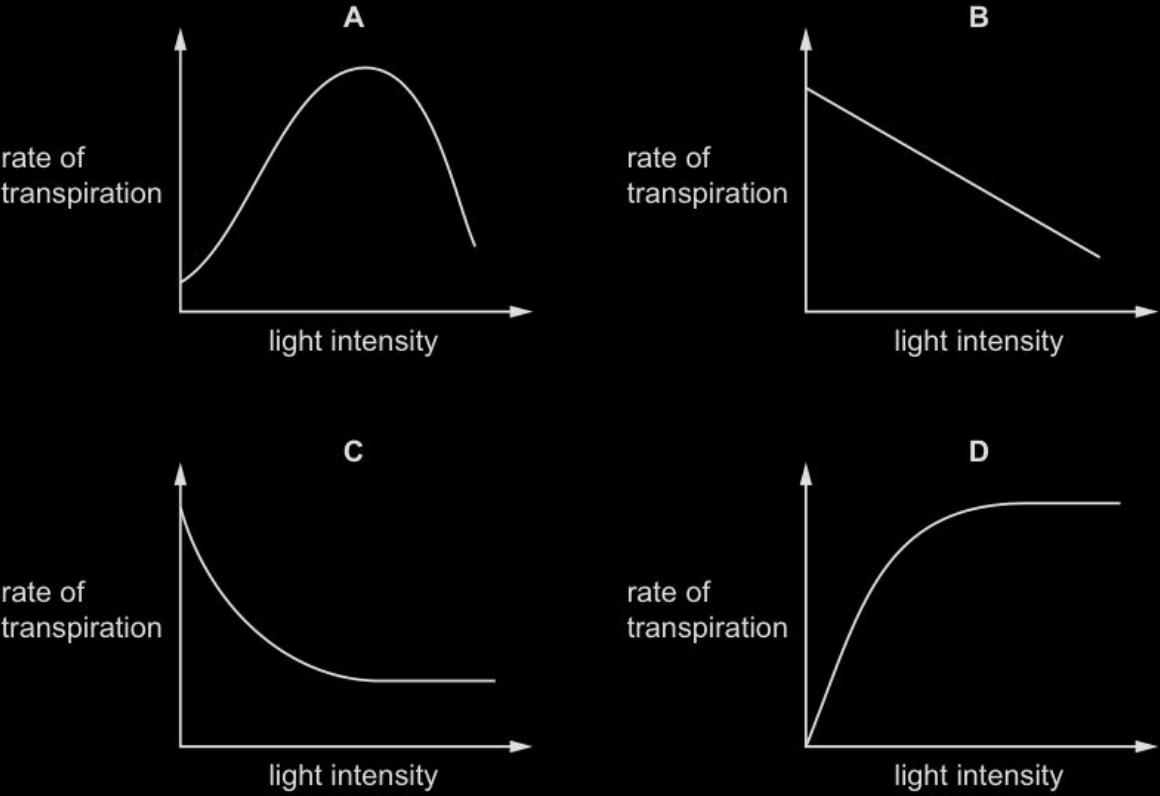
By which pathway does the oxygen diffuse?

- A through the upper epidermis and through the cuticle
- B through the airspaces and through the stomata
- C through the mesophyll and through the phloem
- D through the mesophyll and through the xylem

MS 5090/12/O/N/25 Q8

8	B	1
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10 Which graph shows the effect of light intensity on the rate of transpiration?

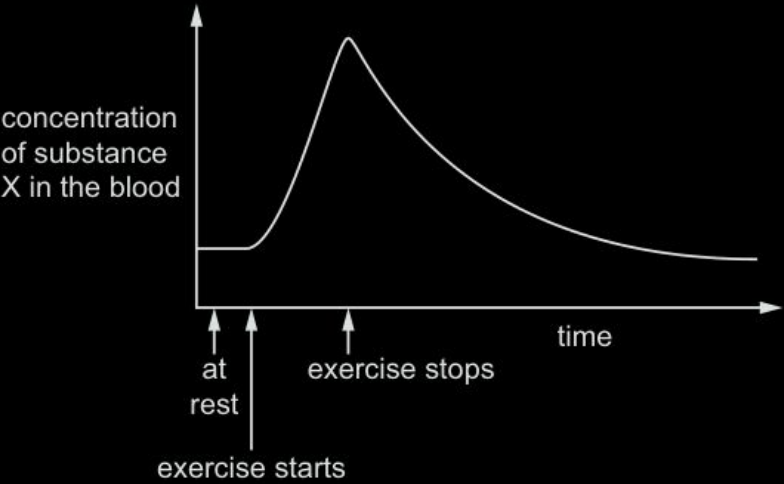


MS 5090/12/O/N/25 Q10

10	D	1
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5090/12/O/N/25 Q14

14 The graph shows the concentration of substance X in a person's blood before, during and after exercise.



What is substance X?

- A alcohol
- B glycogen
- C lactic acid
- D urea

MS 5090/12/O/N/25 Q14

14	C	1
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23 Which term can be used to describe the release of oxygen from plants?

- A breathing
- B excretion
- C respiration
- D transpiration

24 The diagram shows some structures in the skin.

MS 5090/12/O/N/25 Q23

23	B	1
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5090/12/O/N/25 Q35

35 Bread is made from a mixture of flour, sugar, water and yeast.

The mixture is left in a warm place to allow it to rise before being baked.

Which process makes the bread rise?

- A The yeast feeds on the flour and grows.
- B The yeast reproduces to make the loaf bigger.
- C The yeast uses sugar to respire and produces carbon dioxide.
- D The yeast produces alcohol which it excretes.

MS 5090/12/O/N/25 Q35

35	C	1
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